

Evidence Summary - Technology #02

Adaptive Valve Profile for Postural ICP Regulation

External Evidence Sources (5 total):

Source 1: Relationship between valve opening pressure, body

URL: <https://pubmed.ncbi.nlm.nih.gov/15458593>

Excerpt: by M Bergsneider · 2004 · Cited by 77 — This study indicates that very high valve opening pressure settings may be optimal for the initial treatment of normal pressure hydrocephalus .

Source 2: Paradoxical response of intracranial pressure to shunt valve

URL: <https://link.springer.com/article/10.1007/s00701-020-04462-y>

Excerpt: by L D'Antona · 2020 · Cited by 3 — This study describes the effect of adjusting the shunt valve setting on ICP in a cohort of patients with complex symptom management.

Source 3: Understanding Shunt Systems

URL: <https://www.hydroassoc.org/shunt-systems>

Excerpt: Adjustable (programmable) valves regulate the ICP based on a pressure setting, like the fixed pressure valve, but the setting can be adjusted by your doctor

Source 4: Impact of shunt adjustments and body positions

URL: https://www.researchgate.net/publication/263394882_Intracranial_pressure_in_hydrocephalus_Impact_of_shunt_adjustments_and_body_positions

Excerpt: Noninvasive manipulation of programmable shunt valves may allow customization of intracranial pressure (ICP) dynamics in individual shunted patients. Read more

Source 5: Intracranial Pressure Sensor and Valve to Control

URL: <https://www.semanticscholar.org/paper/fcd155029dbc7fa5a82915822f2289055da96164>

Excerpt: This work is working to develop an implantable pressure sensor that will open a valve to relieve the intracranial pressure (ICP)

*Evidence Classification: This dossier contains source-native information, derived engineering analysis, engineering proposals, and unresolved questions.
No evidence has been physically validated.*